



Knowledge elicitation

BSc Introduction to AI
September 13 2022

Eliciting Knowledge from the experts



Experts knowledge

- in their head
- vast amounts
- Tacit (used but hard to describe)
- Valuable

But : reachability is hard

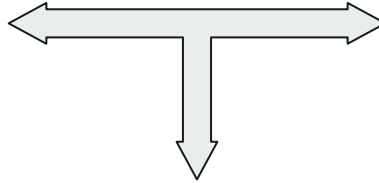
- Experts are very busy
- Each expert doesn't know everything

Need of a (Knowledge) Engineer!

Domain Expert



Knowledge Engineer (yourself!)



Knowledge

Expert Types



Academic

- Regards domain as having a logical structure
- Talks a lot
- Feels a need to present a consistent “story”: teacher
- Often remote from day-to-day problem solving

Practitioner

- Heavily into day-to-day problem solving
- Implicit understanding of the domain
- Emphasis on practical problems and constraints
- Many heuristics

Elicitation Techniques



- Interview
- Self report / protocol analysis
- Laddering
- Concept sorting
- Repertory grids
- Diagram based techniques
- Automated learning techniques
- Induction

More on this next week!

Session Preparation



- Establish goal of the session
- Consider added value for expert
- List relevant questions
- Write down opening and closing statement
- Check recording equipment
 - audio recording is usually sufficient
- Make sure expert is aware of session context
 - goal, duration, follow-up, et cetera

Beginning of the Session : checklist



- ☐ Introduce yourself (if required)
- ☐ Clarify goal and expectations
- ☐ Indicate how the results will be used
- ☐ Ask permission for recording
- ☐ Privacy issues
- ☐ Check whether the expert has some questions left
- ☐ Create as much as possible a mutual trust

During the Session : checklist



- ❑ Avoid suggestive questions
- ❑ Clarify the reasons of a question
- ❑ Phrase questions in terms of probes (e.g, “why ...”)
- ❑ Pay attention to non-verbal aspects
- ❑ Be aware of personal biases
- ❑ Give summaries at intermediate points

End of the Session : checklist



- ☐ Restate the goal of the session
- ☐ Indicate what will be the next steps
- ☐ Make appointments for the next meetings
- ☐ Process interview results ASAP.
- ☐ Organize feedback round with expert
- ☐ Distribute session results

Unstructured interview



- No detailed agenda
- Few constraints
- Delivers diverse, incomplete data
- Used in early stages: feasibility study, knowledge identification
- Optional/Can be skipped
- Useful to establish a common basis with expert
- s/he can talk freely

Structured interview

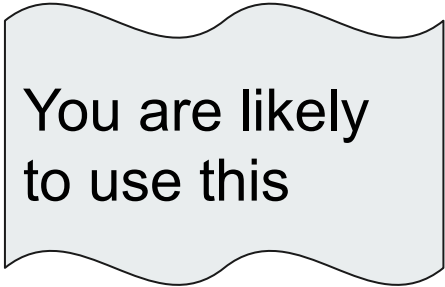


- Knowledge engineer plans and directs the session
- Takes form of provider-elicitor dialogue
- Delivers more focused expertise data
- Often used for “filling in the gaps” in the knowledge base
- knowledge refinement phase
- Also useful at end of knowledge identification or start of knowledge specification
- Always create a transcript

Interview Structure



- Identify a particular sub-task
 - should be relatively small task, e.g. an inference
- Ask expert to identify “rules” used in this task
- Take each rule, and ask when it is useful and when not
- Use fixed set of probes:
 - “Why/how/when would you do that?”
 - “What alternatives are there for this action?”
 - “What if ...?”
 - “Can you tell me more about ..?”



**You are likely
to use this**

Interview Pitfalls



- Experts can only produce what they can verbalize
- Experts seek to justify actions in any way they can
 - “spurious justification”
- You need to: supplement with techniques that observe expertise “in action”
 - e.g. self report

Self Report

- Theoretical basis: cognitive psychology
- Expert performs a task while providing a running commentary
 - Let the expert “think aloud”
- Session protocol is always transcribed
 - input for protocol analysis
- Variations:
 - *shadowing*: one expert performs, a second expert gives a running commentary
 - *retrospection*: provide a commentary after the problem-solving session



**You might use
this**

Self Report : checklist



- Knowledge engineer must be sufficiently acquainted with the domain
- Task selection is crucial
 - only a few problems can be tackled
 - selection typically guided by available scenario's and templates
- Expert should not feel embarrassed
 - consider need for training session

General guidelines (do's and don't's)



- Present problems in a realistic way
- Transcribe sessions as soon as possible
- Avoid long sessions (maximum = 20 minutes)
- Presence of knowledge engineer is important
- Be aware of scope limitations
- Verbalisation may hamper performance
- Knowledge engineer may lack background knowledge to notice distinctions

Other KE techniques



- Observational techniques
 - Observing experts at work
 - Videotaping
 - Can be costly in terms of time/effort
- Limited-information and constrained-processing tasks
 - Either limit the time and/or information available to the expert when performing tasks.
 - E.g Twenty question technique

20 questions technique



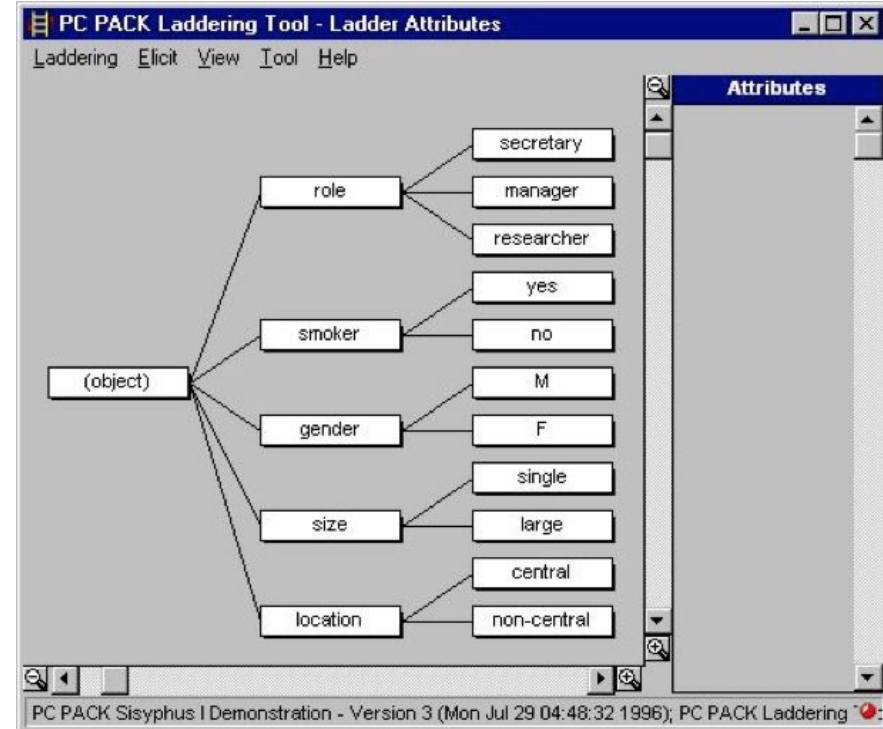
Efficient way of accessing the key information in a domain in a prioritised order:

- a. The Knowledge Engineer thinks of an object/concept domain
- b. The Domain expert asks a series of yes/no questions
- c. Knowledge Engineer notes the questions in the order in which they are asked

Insight into the key aspects, properties or categories and their relative priorities.

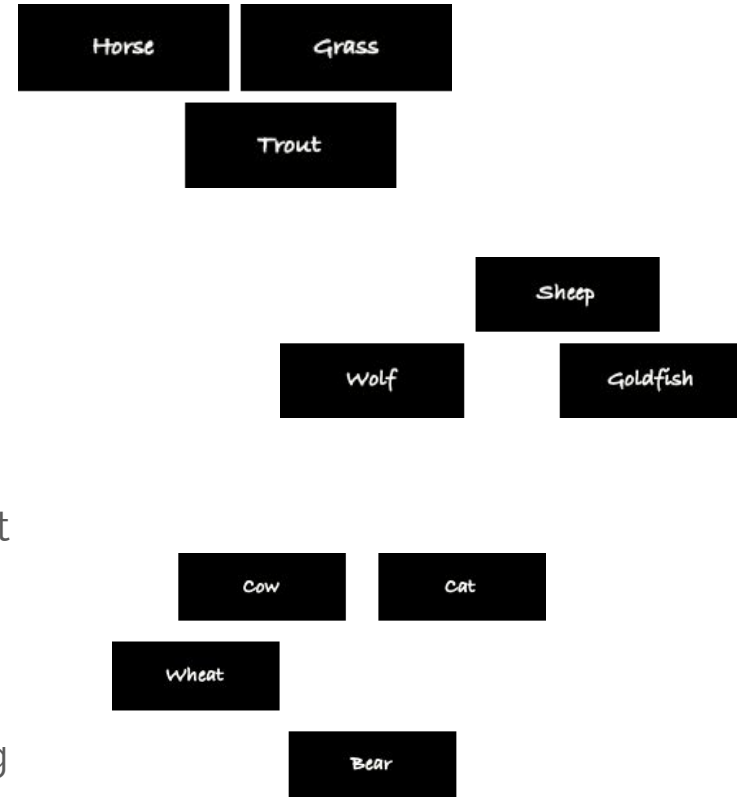
Laddering

- Organizing entities in a hierarchy
- Hierarchies are meant as pre-formal structures
- Nodes can be of any type
 - classes, processes, relations,
- Useful for the initial phases of domain-knowledge structuring
 - in particular knowledge identification
- Can be done by the expert alone

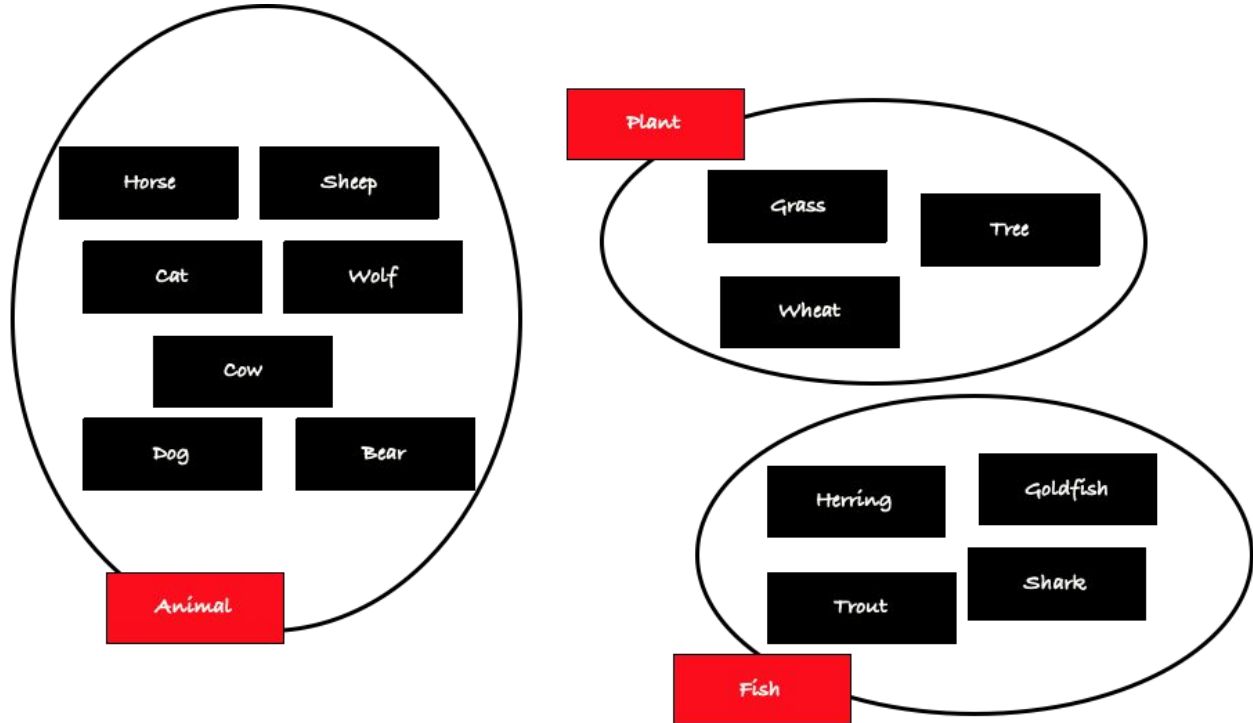


Card Sorting

- Write down each concept/idea on a card
- Organise them in piles
- Link the piles together
 - Identify what the pile represents
 - Record the collections or groupings that have been identified and why these might be considered as groups
 - Do it again, and again
- Each time, note down the results of the sorting



Card Sorting

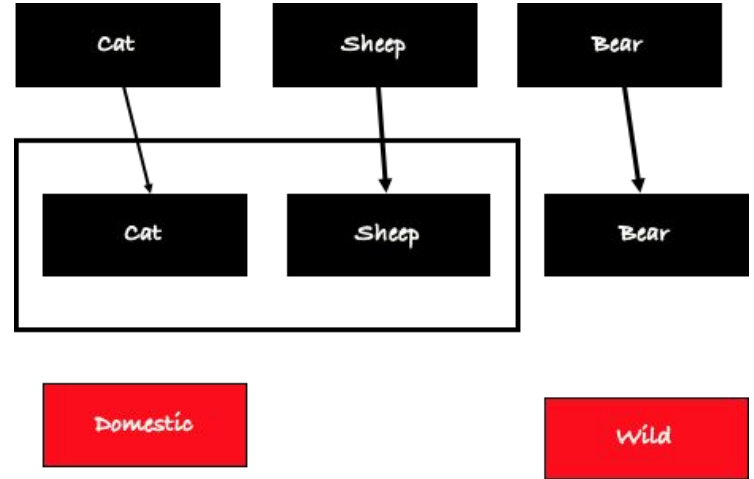
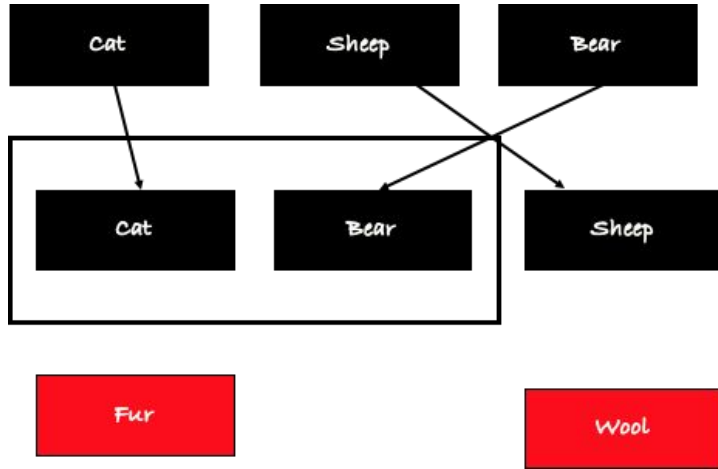


3-Card Trick



- Select 3 cards at random
- Identify which 2 cards are the most similar?
 - Why?
 - What makes them different from the third card?
- Helps to determine the characteristics of our classes
- Picking 3 cards forces us into identifying differences between them
 - There will always be two that are “closer” together
 - Beware of the personal perspective

3-Card Trick



In class exercise



Group by project :

- Which experts do you want to talk?
- What is the goal you want to achieve?
- Which questions will he/she answer?
- When will you meet them?
- Report to class

Next week



Lectures:

- Methodology: AI solutions
- No guest lecture (work on projects)

Tutoring:

- Submit your project idea (research problem)
- Plan your knowledge acquisition and literature review

Assignments:

- 16/09 : Research Problem (Group assignment)
- 19/09 : Reflection on talk by Mark Hoogendoorn (Individual assignment)